



Module Presenter's Manual

for

Data Analytics using Power BI

***Effective from: February, 2025
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	February 2025	New	-

Table of Contents

Sr. No.	Details	Page No.
1.	Introduction	1
2.	Information on Session Allocation	2
3.	Module Deliverables on OnlineVarsity	2
4.	Week-wise Session Schedule	3
5.	Session Coverage	4
6.	Library References	8

1. Introduction

At the end of this course, students will be able to:

- Import, transform, and clean data from various sources using Power BI to ensure accurate and meaningful insights.
- Create interactive reports and dashboards to visualize data effectively for business intelligence and decision-making.
- Apply data modeling techniques to establish relationships between datasets and enhance analysis.
- Use DAX (Data Analysis Expressions) to perform calculations, aggregations, and advanced data analytics.
- Optimize report performance and data refresh schedules to improve efficiency and real-time insights.
- Share and collaborate on Power BI reports using Power BI Service and embed reports in different platforms.

2. Information on Session Allocation

Module	Online Hours	Self-Study Hours
Data Analytics using Power BI	40	12

Throughout this Presenter's Manual, the module, **Data Analytics using Power BI**, will be referred to as **Power BI**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature – Description / Functionality
	Download eBook – Student has the option to download the subject related e-book and read offline
	Practice 4 Me – Student can test and evaluate their understanding of module related topics.
	Work Assignments – Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References – Student can access additional subject related material for reading.
	Show Me How – Student can view a step-wise simulation/demonstration of the module related topics.

4. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2		Day 3	Day 4	
1	Session 1 Power BI – TL1	Session 2 Power BI – TL2		Session 3 Power BI – TL3	Session 4 Power BI – TL4	
2	Session 5 Power BI – TL5	Session 6 Power BI – TL6		Session 7 Power BI – TL7	Session 8 Power BI – TL8	
3	Session 9 Power BI – TL9	Session 10 Power BI – TL10	S1-S3	Session 11 Power BI – TL11	Session 12 Power BI – TL12	
4	Session 13 Power BI – TL13	Session 14 Power BI – TL14		Session 15 Power BI – TL15	Session 16 Power BI – TL16	
5	Session 17 Power BI – TL17	Session 18 Power BI – TL18		Session 19 Power BI – TL19	Session 20 Power BI – TL20	S4-S6

Power BI: Data Analytics using Power BI
TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	Power BI – TL1	All the topics as listed below from Session 1 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 1: Getting Started with Power BI</u> - Define what is Power BI - Define Power BI capabilities - Explain how to Install Power BI Desktop - Illustrate how to navigate the Power BI interface	Unleashing Data Insights with Microsoft Power BI SG-Session 1 XP-Session 1 TG-Session 1
2	Power BI – TL2	The Try It Yourself questions from Session 1 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session.	Unleashing Data Insights with Microsoft Power BI Session 1
3	Power BI – TL3	All the topics as listed below from Session 2 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 2: Exploring Power Query</u> - Define Power Query - Illustrate how to import data with Power Query - Explain the shaping of data with Power Query - Outline the use of Power Query to Fit Analytical Requirements	Unleashing Data Insights with Microsoft Power BI SG-Session 2 XP-Session 2 TG-Session 2
4	Power BI – TL4	The Try It Yourself questions from Session 2 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session.	Unleashing Data Insights with Microsoft Power BI Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
5	Power BI – TL5	All the topics as listed below from Session 3 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 3: Data Management Techniques</u> - Define advanced data extraction - Define data categorization - Explain table transformations - Explain text, numerical, date, and time tools - Illustrate basic QA and profiling tools - Illustrate indexing, grouping, aggregating and pivoting of columns - Explain data preparation techniques in Excel	Unleashing Data Insights with Microsoft Power BI SG-Session 3 XP-Session 3 TG-Session 3
6	Power BI – TL6	The Try It Yourself questions from Session 3 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session.	Unleashing Data Insights with Microsoft Power BI (Onlinevarsity) Session 3
7	Power BI – TL7	All the topics as listed below from Session 4 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 4: Building Data Models</u> - Define and use Data modeling concepts - Illustrate Normalization techniques - Outline efficient data models - Illustrate Star Schema in Power BI	Unleashing Data Insights with Microsoft Power BI SG-Session 4 XP-Session 4 TG-Session 4
8	Power BI – TL8	The Try It Yourself questions from Session 4 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session.	Unleashing Data Insights with Microsoft Power BI Session 4
9	Power BI – TL9	All the topics as listed below from Session 5 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 5: DAX Fundamentals</u> - Define basic DAX Expression - Illustrate the translation of Business Logic - Explain and create basic DAX expressions - Define and convert business logic into DAX formulas	Unleashing Data Insights with Microsoft Power BI SG-Session 5 XP-Session 5 TG-Session 5
10	Power BI – TL10	The Try It Yourself questions from Session 5 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session.	Unleashing Data Insights with Microsoft Power BI Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
	S1-S3	All the Assignments from Sessions 1 to Session 5 of Unleashing Data Insights with Microsoft Power BI course should be solved in these sessions.	Unleashing Data Insights with Microsoft Power BI (Onlinevarsity) Session 1-5
11	Power BI – TL11	All the topics as listed below from Session 6 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 6: Advanced DAX Functions</u> • Define Complex DAX functions • Explain Conditional and logical operations • Elaborate SWITCH function • Classify common text functions • Describe basic date and time functions • Describe CALCULATE function • Describe ALL function • Describe Iterator functions	Unleashing Data Insights with Microsoft Power BI SG-Session 6 XP-Session 6 TG-Session 6
12	Power BI – TL12	The workshop topics from Session 1 to Session 6 of Unleashing Data Insights with Microsoft Power BI course should be covered in this session.	Unleashing Data Insights with Microsoft Power BI (Onlinevarsity) Session 1-6
13	Power BI – TL13	All the topics as listed below from Session 7 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 7: Data Visualization Essentials</u> • Classify and apply visual tools such as cards, charts, tables, and matrices • Illustrate the filtering and mapping of data • Explain and implement slicers • Classify the techniques for drilling up, down, and through data • Describe Advanced chart formatting for enhanced layouts	Unleashing Data Insights with Microsoft Power BI SG-Session 7 XP-Session 7 TG-Session 7
14	Power BI – TL14	The Try It Yourself questions from Session 6 and Session 7 of Unleashing Data Insights with Microsoft Power BI book should be covered in this session.	Unleashing Data Insights with Microsoft Power BI Session 7

Session No.	Session Title	Session Details	Deliverables' Mapping
15	Power BI – TL15	All the topics as listed below from Session 8 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 8: Advanced Reporting with AI</u> - Define and apply AI tools in Power BI - Illustrate Advanced data visualization - Classify and design datasets using ChatGPT - Illustrate and Execute Query based ChatGPT data - Illustrate handling of missing data	Unleashing Data Insights with Microsoft Power BI SG-Session 8 XP-Session 8 TG-Session 8
16	Power BI – TL16	The Try It Yourself questions from Session 8 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session.	Unleashing Data Insights with Microsoft Power BI Session 8
17	Power BI – TL17	All the topics as listed below from Session 9 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 9: Advanced Dashboards and Analysis</u> - Illustrate dashboard interactivity and design - Explain data navigation and analysis - Illustrate the technique of visualization integration dynamic reporting with bookmarks and tooltips - Illustrate advanced dashboards	Unleashing Data Insights with Microsoft Power BI SG-Session 9 XP-Session 9 TG-Session 9
18	Power BI – TL18	All the topics as listed below from Session 10 of <i>Unleashing Data Insights with Microsoft Power BI</i> book should be covered in this session. <u>Session 10: Power BI Service, Collaboration, and Optimization</u> - Explain uploading and refreshing Power BI reports - Illustrate publishing and sharing reports - Illustrate report optimization using Power BI tools - Explain visual optimization	Unleashing Data Insights with Microsoft Power BI SG-Session 10 XP-Session 10 TG-Session 10
19	Power BI – TL19	The workshop topics from Session 7 to Session 10 of <i>Unleashing Data Insights with Microsoft Power BI</i> course should be covered in this session.	Unleashing Data Insights with Microsoft Power BI (Onlinevarsity) Session 7-10
20	Power BI – TL20	The Try It Yourself questions from Session 9 and Session 10 of <i>Unleashing Data Insights with</i>	Unleashing Data Insights with Microsoft Power BI Session 10

Session No.	Session Title	Session Details	Deliverables' Mapping
		Microsoft Power BI book should be covered in this session.	
	S4-S6	All the Assignments from Sessions 6 to Session 10 of Unleashing Data Insights with Microsoft Power BI course should be solved in these sessions.	Unleashing Data Insights with Microsoft Power BI (Onlinevarsity) Session 6-10

Post Class Activity: Students should solve the given Assignments and submit them to the trainers. They can go through the other components of the course such as References.

6. Library References

➤ Mastering Microsoft Power BI by Brett Powell
➤ Microsoft Power BI Cookbook by Greg Deckler
➤ Analyzing Data with Power BI and Power Pivot for Excel by Alberto Ferrari, Marco Russo

~~ End of Document ~~

For Aptech Centre Use Only